

K4 Breakthrough Thoughts

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Preface

In March, 2013, I discovered that K4 uses multiple steps (as many had suspected), and decoded one of the steps. The first step uses cryptography that I am sure would not surprise those that solved K1 through K3.

On May 20, 2013, 2 1/2 years after Jim Sanborn released the BERLIN clue, I announced this to the Kryptos Yahoo Group.

The 97 characters in the next step of K4 are (split into two lines for readability):

PGZFGTEZPFCYGFLLJLZGSOJAILWDCCNZFHSHQHIVZKUCGOZAAF
JEVHOCZVSNHYEHBBERLINPEEKRRRRHBUXJOPYFFTXBMMQMOFQS

I do not have a background in cryptography, so I am just a novice (who, through some combination of luck or skill, figured out part of K4).

Following are some thoughts on how to proceed from here.

Thought #1: A Mask

One of the most noticeable features of the partial decoding of K4 is the presence of BERLIN, which is generally assumed to be in the final decoding.

This leads to two possibilities: [1] The final text does not include BERLIN (with the word Berlin having randomly appeared in the middle of the decoding steps), or [2] Berlin does appear in the final decoding.

I find #1 unlikely. Having looked at perhaps thousands of incorrect decodings of K4, many have short English words in them. However, getting a 6 letter word is fairly rare - especially one that has such a clear relation to Kryptos (e.g. with the Berlin wall monument at the CIA headquarters near Kryptos). And many people would call "foul" if the final text of K4 did not indeed have BERLIN where the clue suggested it would be.

So if we assume that BERLIN is in the final decoded text where it appears in this partial solution, I think that there must be a mask of some sort involved (something that allows further decoding of some characters, but not others, essentially allowing the BERLIN letters to bypass further decoding).

The most obvious mask I could see was the morse code in Kryptos. My thought was to put the morse code together in one string, and where the extra letters 'e' appeared no encoding would be done.

As an example, let's say the the ciphertext is "BERLINPEEKRRRHB", mask is "eeeeeeINVISIBLE" (with any 'e' outside a word meaning 'do not decode', anything else meaning 'encode'), and the encoding is a simple Caesar cipher (add 1 to the letter, so 'A' becomes 'B'). In this case, it would decode to BERLINQFFLSSSIC. The mask allows further decoding while still letting BERLIN through.

We would have:

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ciphertext: ...BERLINPEEKRRRHB...
mask:       ...eeeeeeINVISIBLE...
plaintext:  ...BERLINQFFLSSSIC...
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Of course, this is simplistic (e.g. the 'RRR' likely requires multiple alphabets), but shows how a mask might work.

I did try as many possible variations of the morse code as I could (as it it not clear exactly the order that morse code snippets would go in), without luck.

If a mask is used, and it lets BERLIN through, the mask would need to be at least 97 characters long (or repeated to that length). Also, it would need to have at least 6 "do not decode this letter" marks in it in a row (like the 'e's in the morse code) to allow BERLIN through.

Thought #2: A 14-Character Key

Running some statistics on the partial solution to K4 (specifically, the periodic index of coincidence) reveals that there may be a 14-character key involved. I calculated them as follows (for key lengths of 2 through 21):

2,0.0399	3,0.0355	4,0.0372	5,0.0382
6,0.0355	7,0.0434	8,0.0389	9,0.0438
10,0.0394	11,0.0332	12,0.0321	13,0.0394
14, 0.0578	15,0.0292	16,0.0292	17,0.0275
18,0.0426	19,0.0193	20,0.0250	21,0.0317

Here, you can see that the index of coincidence is highest for a key length of 14. Unfortunately, I am not experienced enough in ciphers to know how likely a key length of 14 is given this data (especially given the short length of ciphertext and the possibility of a mask). If a periodic cipher is used with a key (such as Vigenere), however, the data suggests a 14-character key.

Further, the duplicated letters in the partial solution of K4 are "ZF" (at positions 2 and 30), and "HB" (at positions 62 and 76). The difference between the "ZF" positions is 28 (a multiple of 14), and the difference between the "HB" positions is 14. I believe this helps increase the chances that a 14-character period is used.

So where could one find a 14-character key? Using a word that long is very unlikely; my guess would be it has to do with there being 14 rows on the panels.

Breaking down the partial solution into 14-character segments results in:

```
PGZFGTEZPFCYGF
LJLZGSOJAILWDC
CNZFHSHQHIVZKU
CGOZAAFJEVHOCZ
VSNHYEHBERLINP
EEKRRRHBUXJOPY
FFTxBMMQMOFQS
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The idea of a 14-character key comes from the PLCCVEF (first column) using one alphabet, GJNGSEF (second column) using another, and so forth. That could be a Vigenere cipher, or something completely different.

However, one would still need to account for BERLIN staying where it is (assuming, of course, it does stay there). The repeated HB suggests that the 'H' before BERLIN is indeed plaintext, and that the second HB is plaintext as well.

I tried some common methods of attacking the partial decoding with 14-letter keys, but none worked for me. That could be due to some letters not being decoded (e.g. BERLIN staying as-is with a mask), or it could be due to multiple steps being required.

Thought 3: Two Halves; K1+K2 Alignment

Something that is quite apparent (to me, at least) is that the first half of K4 seems somehow *different* than the second half of K4. It's hard to place in words how it seems different. And it may be more than just the halves. For example, splitting the partial K4 solution into quarters shows an IC (Index of Coincidence) of the 1st and 3rd quarters being identical, with the 2nd and 4th quarters lower.

This may just be due to the short length of K4. Statistics work better and better on longer and longer lengths of ciphertext.

The section after this has some of the statistics.

Next, with the idea of Kryptos perhaps being a palimpsest, I noted that K1 is 63 bytes, with K2 starting at the 64th character of the first panel. Interestingly, BERLIN starts at the 64th position of K4.

Encoded K1 (bold) and start of K2:	Partially decoded K4 (BERLIN in bold):
EMUFPHZLRFAXYUSDJKZLDKRNSHGNFIVJ	PGZFGTEZPFCYGFLLJLZGSOJAILWDCCNZF
YQTQUXQBQVYUULLTREVJYQTMKYRDMFD	HSHQHIVZKUCGOZAAFJEVHOCZVSNHYEH
VFPJUDEEHZWETZYVGWHKKQETGFQJNCE	BERLIN PEEKRRRHBUXJOPYFFTXBMMQMO
GGW	FQS

This gives the possibility that the first half of K4 is somehow related to K1, with the second half of K4 is somehow related to K2.

My hunch is that K4 is indeed split into two (or more) sections somehow, but that the positioning of BERLIN starting at the 64th position (correlating to the start of K2) is just a coincidence.

Thought 4: Statistics - Index of Coincidence

Here is the Index of Coincidence for the encoded and decoded K1, K2, K3, and K4. As a reminder, random letters would tend to be closer to 0.038, with standard English words tending to hover around 0.066.

K1:	0.038	K1 Decoded:	<u>0.067</u>
K2:	0.045	K2 Decoded:	<u>0.066</u>
K3:	0.066	K3 Decoded:	<u>0.066</u>
K4:	0.036		

K4 Partial Solution:

All 97 chars:	0.037		
1 st 48 chars:	0.045	2 nd 48:	0.039
1 st 24 chars:	0.054	2 st 24 chars:	0.036
3 rd 24 chars:	0.054	4 th 24 chars:	0.047

From the IC, it is pretty clear that we are not dealing solely with transposition (I.E. at least some of these 97 letters need to be changed into other letters).

As was mentioned in the previous section, statistics work better when you have more letters to work with. The statistical breakdown of the smaller segments could easily just be "noise".

Thought 5: Length

There are 97 letters in K4. That poses a problem for some ciphers, as 97 is a prime number (no two numbers multiplied together result in 97).

If a cipher is involved that requires the use of a non-prime number of characters, I would find 98 characters to be the most likely (perhaps including the '?' at the end of K3, or perhaps a null character). 98 would work into 7 14-character lines, which fits in nicely with the probability of a 14-character key. It also fits in nicely with the solution to K3, where Jim Sanborn's chart has 7 lines.